

练习十八 热力学基础（三）

1. 等压: $\frac{1}{2}RT_0$

2. $467K$; $234K$

3. (3)

4. (3)

5. (1) $1 \rightarrow 2$ 等温: $p_1V_1 = p_2V_2$ $p_2 = \frac{V_1}{V_2} p_1 = 5atm$

$2 \rightarrow 3$ 绝热: $T_1V_2^{\gamma-1} = T_2V_3^{\gamma-1}$ $V_3 = \left(\frac{T_1}{T_2}\right)^{\frac{1}{\gamma-1}} V_2 = 48.8 \times 10^{-3} m^3$

$p_2V_2^\gamma = p_3V_3^\gamma$ $p_3 = \left(\frac{V_2}{V_3}\right)^\gamma p_2 = 1.43atm$

$4 \rightarrow 1$ 绝热: $p_4^{\gamma-1}T_2^{-\gamma} = p_1^{\gamma-1}T_1^{-\gamma}$ $p_4 = \left(\frac{T_1}{T_2}\right)^{\frac{\gamma}{\gamma-1}} p_1 = 2.87atm$

$3 \rightarrow 4$ 等温: $p_3V_3 = p_4V_4$ $V_4 = \left(\frac{p_3}{p_4}\right) V_3 = 24.4 \times 10^{-3} m^3$

(2) $W = Q_1 - |Q_2| = \frac{m}{M} R \left(T_1 \ln \frac{V_2}{V_1} - T_2 \ln \frac{V_3}{V_4} \right) = (p_1V_1 - p_3V_3) \ln 2 = 2.1 \times 10^3 J$

(3) $\eta = 1 - \frac{T_2}{T_1} = 30\%$

6. $W = -\Delta E = \mu \frac{i}{2} R(T_0 - T) = \frac{i}{2} (p_0V_0 - pV)$

又 $\gamma = \frac{C_p}{C_v} = \frac{i+2}{i} = 1 + \frac{2}{i} \Rightarrow \frac{i}{2} = \frac{1}{\gamma-1}$ $W = \frac{p_0V_0 - pV}{\gamma-1}$